A maximum entropy perspective reveals deviations from steady state during active diversification

Abstract

Aim

Ecosystems are rarely at steady state, yet most eco-evolutionary theory and modeling that predict universal biodiversity patterns assumes they are. Here, we test whether and how eco-evolutionary dynamics themselves drive departures from steady state.

Location

Hawaiian Archipelago.

Time Period

Current.

Major Taxa Studied

Arthropods.

Methods

We combined arthropod community data from the geologic chronosequence of the Hawaiian Archipelago with the Maximum Entropy Theory of Ecology (METE). The chronosequence of the Hawaiian Archipelago offers insights into eco-evolutionary processes because ecosystems growing on different aged substrates offer snapshots of community assembly with different histories. METE is a minimalist steady-state framework that simultaneously predicts species abundance distributions (SAD) and individual metabolic rate distributions (IPD). We quantify departures from steady state as the deviation of observed SADs and IPDs from there theoretical predictions under METE. We then test how those deviations are predicted by position along the chronosequence, proportion of non-native species, and β-diversity using metaregression. We also use null model analysis to understand how the contribution of non-native species to β-diversity changes across the chronosequence.

Results

Deviations from METE peak at geologically middle-aged sites (150 Kya–1.4 Mya), consistent with active adaptive radiation pushing communities away from statistical steady state. β-diversity, which also peaks at middle-aged sites, robustly predicts deviations from METE across all sites, while the proportion of non-native species predicts deviations only after excluding the geologically youngest site. Partitioning β-diversity between native and non-native species resolves this discrepancy. At the youngest site, non-native species are distributed homogeneously and do not elevate β-diversity despite their high proportional representation.

Main conclusions

These results are consistent with a trajectory from young, dispersal-assembled communities near statistical steady state, through an eco-evolutionary non-steady-state transition driven by diversification, to a new stable steady state at the oldest sites. Our findings suggest that periods of active diversification create windows of ecological instability that may facilitate biological invasion.

Introduction

Biodiversity patterns are bound by two seemingly contradictory states: they show striking consistency in form across systems, and yet also must emerge in spite of, or because of, the constant flux and dynamic change of ecology and evolution. The near universality of biodiversity patterns includes the scaling of species richness with area (Rosenzweig *et al.*, 1996; Harte *et al.*, 2009), the ubiquity of rare species and scarcity of common species (Hubbell, 2001; Harte, 2011), and the highly skewed distribution of biomass or metabolic resource use across

organisms (Brown, 1995). Debate has persisted for over a century as to whether these consistent patterns form because of ecological and evolutionary drivers that force systems into these configurations, or are largely the outcome of chance acting on large systems of constituent parts (Clements, 1916; Gleason, 1926; Fisher *et al.*, 1943; Preston, 1948; MacArthur & Wilson, 1967; Rosenzweig *et al.*, 1996; Hubbell, 2001; Adler *et al.*, 2007; Chisholm & Pacala, 2010; Harte, 2011; Diaz *et al.*, 2021). The deterministic view gives rise to niche-based theories of coexistence (Chesson, 2000; Barabás *et al.*, 2018), while the stochastic view has produced a number of neutral-like models (Hubbell, 2001; Harte, 2011).

Both deterministic and stochastic perspectives on the generation of universal biodiversity patterns most often assume an ecosystem is at "equilibrium" or steady state (Chesson, 2000; Hubbell, 2001; Harte, 2011; Barabás *et al.*, 2018). However, ecosystems are known to be dynamic both through intrinsic eco-evolutionary processes (Gillespie, 2004; Knope *et al.*, 2012; Rominger *et al.*, 2016) and due to recent anthropogenic perturbation (Chen *et al.*, 2011; Graham *et al.*, 2017; Barton *et al.*, 2021; Garcia *et al.*, 2022). Whether universal patterns are insensitive to non-steady-state processes or are a result of those processes remains un-addressed. To better understand the interaction of non-steady-state processes and universal patterns, we combine data (Gruner, 2007) from the highly dynamic forest ecosystems of the Hawaiian Archipelago (Wagner & Funk, 1995) with the intentionally minimalist steady state predictions of the Maximum Entropy Theory of Ecology (Harte, 2011).

The Hawaiian Archipelago Chronosequence

The Hawaiian Archipelago holds numerous examples of adaptive radiation (Baldwin & Sanderson, 1998; Gillespie, 2004; Givnish *et al.*, 2009; Lerner *et al.*, 2011; Knope *et al.*, 2012; Bennett & O'Grady, 2013; Magnacca & Price, 2015). Those remarkable examples of diversification along with the geographic isolation and chronological geology of the archipelago have made it a focal point for the study of eco-evolutionary processes (Wagner & Funk, 1995; Chadwick *et al.*, 1999; Price & Wagner, 2004; Gillespie & Clague, 2009; Barton *et al.*, 2021). The manner in which the islands formed in chronological sequence Figure 1 has allowed for the study of ecosystems in different stages of eco-evolutionary assembly (Gillespie, 2004; Vitousek, 2004; Rominger *et al.*, 2016) by considering ecosystems on different aged substrates to be snapshots of eco-evolutionary processes at different time points. Age is a key axis in hotspot archipelagos because it affects diversification both directly and indirectly (Whittaker *et al.*, 2008; Gillespie & Baldwin, 2009; Borregaard *et al.*, 2017; **lim2017?**). Directly, speciation takes time to complete, typically on the order of 1 My for Hawaiian arthropods (Hormiga *et al.*, 2003; Gillespie, 2004; Bennett & O'Grady, 2013). Indirectly, island ontogeny produces predictable changes in area and topographic complexity as islands grow and then erode, which in turn affect speciation and extinction dynamics (Whittaker *et al.*, 2008; Borregaard *et al.*, 2017). Lineages also tend to colonize the archipelago from older to younger islands (the progression rule) (e.g., Wagner & Funk, 1995; Gillespie & Clague, 2009), so age also orders the source populations available to any given site.

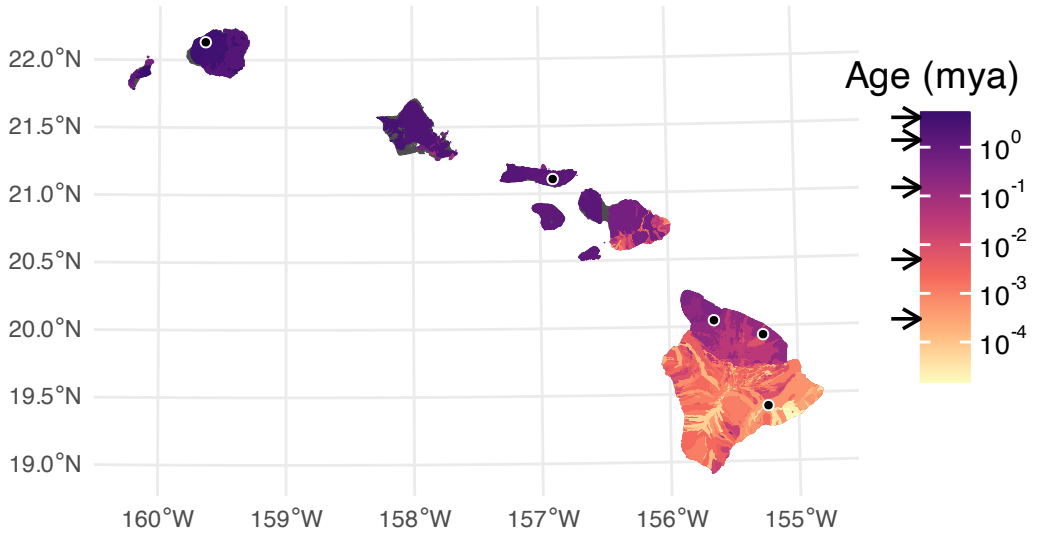


Figure 1: Chronosequence of the Hawaiian Archipelago showing the oldest flows to the north west on Kauaʻi and youngest flows to the south east on Hawaiʻi. Substrate age data from Sherrod *et al.* (2021). The five sites are indicated with dots on the map and arrows indicating their flow ages on the legend. From oldest to youngest sites are: Alakaʻi (Kauaʻi: 4.1 Mya), Kamakou (Molokaʻi: 1.4 Mya), Kohala, Laupāhoehoe, Kīlauea (Hawaiʻi: 0.5 Mya, 150 Kya, 300 ya).

The Hawaiian Archipelago has also absorbed pervasive anthropogenic changes to its biodiversity, with a rapidly expanded human land-use footprint and markedly accelerated introduction of non-native species following Western contact and settler colonialism (Graham *et al.*, 2017; Gon *et al.*, 2018). As such the Hawaiian Archipelago has produced numerous studies of rare and endangered species (Magnacca, 2007; Pratt *et al.*, 2009; Régnier *et al.*, 2015) and the functions of non-native species in ecosystems (Ostertag *et al.*, 2009; Barton *et al.*, 2021).

As a result of the combination of ancient and recent processes, arthropods assemblages across the Hawaiian Archipelago are now composed of members of adaptive radiations (Gillespie, 2004; Bennett & O'Grady, 2013; Magnacca & Price, 2015), many rare species (Magnacca, 2007; Magnacca & Price, 2015), and introduced species (Nishida, 1992). Given the abundance and diversity of arthropods, they also present a unique opportunity to use ecological theory to study how biodiversity conforms to or deviates from steady state.

The Maximum Entropy Theory of Ecology

The Maximum Entropy Theory of Ecology [METE; Harte (2011)] is a particularly useful tool for studying steady state biodiversity patterns because it produces multiple testable predictions, including the distribution of abundances across species and the distribution of metabolic rates across individuals, making falsifications of the theory more informative (McGill, 2003).

The species abundance distribution (SAD) and distribution of metabolic rates across individuals (IPD for individual power distribution; power because metabolism is energy per time) contain complementary and overlapping information about an assemblage. The SAD contains information about species diversity and encodes signals of different demographic processes (Engen & Lande, 1996). The IPD contains information about how energetic resources are partitioned across individuals which encodes information both about the life history strategies of different lineages and the efficiency with which organisms can acquire useful resources from their environments for reproduction (Brown *et al.*, 1993; Burger *et al.*, 2019). This link to reproduction interconnects abundance and metabolic rate distributions (Damuth, 1987; Brown *et al.*, 2004).

METE arrives at some similar predictions to other null or neutral models of biodiversity such as the Unified Neutral Theory (Hubbell, 2001) but instead of being limited to a single neutral formulation, METE is grounded in the principles of statistical mechanics (Jaynes, 1957; Harte, 2011) drawing from the probabilistic properties of large, randomly assembled systems. METE posits that biodiversity patterns are statistical outcomes from a few simple state variables, independently of the specific biological mechanisms governing the values of the state variables. The state variables of METE are total number of species S_0, total number of individuals N_0, and summed metabolic rate across all individuals E_0 (Harte, 2011). Thus, its predictions represent a community in statistical steady state. The steady state assumption holds so long as the governing biological mechanisms do not introduce information beyond what the state variables themselves capture. Examples of such additional information could be rapid environmental change leading to mismatched niches with local environments, rapid diversification, or biological invasion. Under such scenarios the simple state variables of S_0, N_0, and E_0 would not likely constitute enough information to yield accurate predictions for broad biodiversity patterns such as the species abundance distribution or distribution of metabolic rates across individuals. Thus finding deviations between observed patterns of abundance and metabolic rate distributions compared to their METE predictions allows us to identify ecological systems out of statistical steady state (Harte, 2011; Rominger *et al.*, 2016).

METE across the chronosequence

In this study of arthropods communities across the chronosequence, we find consistent deviations from steady state at middle-aged sites. We therefore ask: what could be the drivers of non-steady state at these middle-aged sites? Our motivating hypothesis is that steady state is reached quickly at the youngest site through dispersal-based assembly, transitioning to non-steady state assembly at middle-aged sites driven by evolutionary diversification that culminates at the oldest site in a new steady state. With the data at hand we can begin to chip away at that larger hypothesis by investigating two corollary hypotheses: 1) that peak diversification at middle aged sites leaves a signature of increased β-diversity (Economo & Keitt, 2008; Graham & Fine, 2008; Hofmann *et al.*, 2019) and β-diversity will predict deviations from METE; and 2) the spread of non-native species, an inevitable confounding factor in the

Hawaiian Archipelago, disrupts eco-evolutionary assembly and results in non-steady-state and thus departures from METE.

Deviations from METE are indeed predicted by within site β-diversity which peaks at middle aged sites, consistent with rapid diversification processes. Deviations are also partially predicted by proportion of non-native species at a site, except that the proportion of non-native species is high at the youngest site while this site also shows some of the lowest deviations from METE. The conundrum of the youngest site is resolved by considering the contributions of native and non-native species to β-diversity. The high contribution of native species to β-diversity at middle-aged sites again supports the possibility of rapid diversification playing a role in departures from steady state. More work connecting deviations from steady state to ecological and evolutionary drivers will be needed to further test these possibilities.

Methods

Reproducibility and data preparation

A custom R package, *HawaiiArthMETE* accompanies this paper and can be used to reproduce all results. Supplementary Section S1 details how to access and install *HawaiiArthMETE*. All analysis scripts and data analyzed are available via the package.

Arthropod data come from the original publication by Gruner (2007), a study that collected 17024 specimens across the chronosequence Figure 1. The five study sites in Gruner (2007) were standardized to be at very similar elevation and climatically homogeneous (Vitousek, 2004), seeking to capture differences in arthropod assemblages due to substrate age and not other abiotic factors. Arthropod assembledges were collected by canopy fogging from the endemic ʻōhiʻa lehua (*Metrosideros polymorpha* Gaud. Myrtaceae), which is the dominant tree in mesic to wet, mid-elevation forests the Hawaiian Archipelago. Each site contained between 8 and 11 replicate samples. Replicate samples were selected to be representative of the site and not introduce confounding variation by selecting trees of modal size (size being representative of local climate and water availability) and within forest areas sharing a common ʻōhiʻa lehua dominated canopy. Each specimen was identified to the species level (morpho species were used in 31.56% of specimens). Specimen lengths were measured to the nearest millimeter.

METE is constrained by total metabolic rate (E_0) and predicts a metabolic rate distribution, thus we need to estimate metabolic rate from these data. We use a 3/4 scaling rule (Brown *et al.*, 2004) to calculate metabolic rate as $B = M^{3/4}$ where M is mass. Because METE normalizes metabolic rates such that the lowest possible rate is 1 we can safely ignore units and any scaling constants (e.g. b_0 in $B = b_0 M^{3/4}$).

This leaves the estimation of mass from length, taxonomy, and life stage. Gruner (2007) used taxon- and stage- specific length-to-biomass allometric scaling relationships (Gruner, 2003) to approximate mass as the expected value of a log-log regression

$$\text{mass} = e^{\widehat{\log(\text{mass})}}$$
$$\widehat{\log(\text{mass})} = \log(a) + b \cdot \log(\text{length})$$

where a and b are estimated from data collected by Gruner (2003) and vary by taxon and stage.

Using these expected value masses based on length, the latter measured to the nearest millimeter, yields metabolic rate approximations that are artificially discretized (see the appendix "Sampling biomass estimates from allometric equations"). These discretizations in turn lead to spurious conclusions about the predictiveness of METE because METE assumes metabolic rates are continuous (see Supplementary Section S6 and sfig{fig-ipd-both}).

Therefore, instead of estimating mass as the expected value from an allometric relationship we sample masses as random draws from the full model relating mass to length

$$\text{mass} = e^{y}$$
$$y \sim \mathcal{N}(\widehat{\log(\text{mass})},\ \sigma)$$
$$\widehat{\log(\text{mass})} = \log(a) + b \cdot \log(\text{length})$$

where σ is the taxon- and stage-specific standard deviation of residuals. When the data set from Gruner (2007) reports only biomass and not length, we use reported biomass for the calculation of $\widehat{\log(\text{mass})}$ and use σ estimated from samples with length measurements, or set $\sigma = 0$ if there are no appropriate length measurements. The full procedure for sampling masses is detailed in the appendix "Sampling biomass estimates from allometric equations" and the consequences of using random draws versus expected values on the METE analysis is thoroughly investigated in Supplementary Section S6. We find that using random draws for mass estimates is largely similar to using expected values, and our overall conclusions are unchanged, but using draws avoids both spurious deviations from and conformations to METE that would be introduced by using expected values of mass.

The focus on metabolic rates, and thus estimated biomass, is necessary both because METE is constrained by total metabolism (E_0) and because back-converting the METE-predicted metabolic rate distribution to a length distribution (the variable directly measured in our data set) is not possible in practice. Back-converting length from metabolism rates means first inverting taxon and stage-specific allometric equations. That first step is easily done, but to apply those inverted taxon and stage-specific allometric equations, we need to know the taxonomy and life stages are for each organism. METE is not given any information about taxonomy (only number of species S_0, not identify) and life stage. Therefore METE cannot tell us what the life stage and taxonomy are for any organism of a given metabolism, and thus we cannot correctly apply the inverted allometric equations to derive a METE-predicted length distribution.

METE analysis

We produce METE predictions for each sample within a site, including both native and non-native species in the calcualtion of state variables, SADs, and IPDs. We use the *meteR* R package (Rominger & Merow, 2017) for all METE calculations. We compare METE predictions to observed SAD and IPD patterns using a z^2-value. The use of z^2-values is discussed in Rominger & Merow (2017), but briefly: the z^2-value compares the log likelihood of the data given the METE model to the log likelihood of a hypothetical data set drawn from the METE model itself. This comparison is done many times and the actual z^2-value is calculated as

$$z^2 = \left(\frac{\log(\mathcal{L}(\text{data} \mid \text{METE})) - \hat{\mu}_{\log \mathcal{L}_{\text{METE}}}}{\hat{\sigma}_{\log \mathcal{L}_{\text{METE}}}} \right)^2$$

where $\hat{\mu}_{\log \mathcal{L}_{\text{METE}}}$ is the mean of the log likelihood of data drawn from METE itself, and $\hat{\sigma}_{\log \mathcal{L}_{\text{METE}}}$ is the standard deviation of the log likelihood of data drawn from METE itself.

If METE is truly correct in describing the data, z^2 will asymptotically have a χ^2 distribution with $d.f. = 1$ (Rominger & Merow, 2017). If METE does not truly describe the data, the z^2-value is likely to be large. When visualizing z^2-values we show the 0.05 significance critical value cut off (3.841). This cut off marks the z^2-value above which the observed log likelihood would have a ≤ 0.05 probability of being observed if the data were directly sampled from METE. We emphasize, however, that we use z^2 primarily as a standardized, continuous measure of departure from METE: larger values indicate greater departure and smaller values indicate closer conformity, and we are interested in how that departure varies across the chronosequence rather than in a binary decision about METE. Failing to exceed the critical value is not the same as accepting METE. A z^2-value below the cut off means only that METE is more plausible than if the z^2 value were above the cut off, not that METE perfectly describes the data. That z^2 measures real departure from METE regardless of proximity to the critical value is shown directly by relating z^2 to the differences between observed and predicted rarity and dominance (see below).

To further understand differences between METE predictions and observed SADs and IPDs, beyond what the z^2 statistic alone can tell us, we compare the extrema of the observed and predicted distributions. These extrema can be especially informative about the nature of deviations (Harte, 2011). For the SAD we compare rarity as the log ratio of observed proportion of singletons to the predicted proportion of singletons, and we compare dominance as log ratio of observed total abundance of the top three most abundant species to the predicted total abundance of the top three most abundant species. For the IPD we compare metrics similar to rarity and dominance: the log ratio of observed proportion of individuals with a scaled metabolism of less than 3 to its prediction, and the log ratio of the observed sum of metabolic rates for the top 10 individuals to its prediction.

Statistical analysis of deviations from METE and potential explanatory variables

We first evaluate how deviations from METE, as measured by z^2-values, vary across the chronosequence. We then test for relationships between z^2-values and proportion of non-native species at a site as well as within site β-diversity.

For all these analyses we fit linear models predicting log-transformed z^2-values responding to each explanatory variable: substrate age, proportion non-native species, and within site β-diversity. Log-transforming z^2-values brings them closer to the normal assumptions of a linear model. The need for log-transformation is not surprising: while z^2-values will be χ^2 distributed if METE is true, deviations from METE add more probability to larger values, and taken together will lead to a right-skewed distribution of z^2-values. Log-transformation brings right-skewed random variables closer to normal.

When modeling deviations from METE, we have two statistical challenges to address: (1) samples within sites could represent pseudo replicates if each was given equal weight in a fixed effects model. However, (2) a mixed-effects model (one method to account for otherwise pseudo replication) presents its own challenges: because explanatory variables are site-level characteristics with only one unique measurement per site and there are only five sites, site-level random effects will be poorly estimated leading to poorly estimated uncertainty around our effect sizes of explanatory variables.

To address these challenges we aggregate within site replicates into single means and variances per site. We then model these site-level means and variances using fixed effects meta-regression. Compared to models that would use all data points (i.e., not summarized as means and variances) this is a conservative approach and could run the risk of being under-powered because while we have 48 samples, we only have 5 sites. Meta-regression was originally developed for meta-analysis applications and produces a linear model using maximum likelihood, or (equivalently under normallity assumptions) weighted least squares where the weight of each site mean is the inverse of its site variance. Because these methods are well established in meta-analysis, we use the meta-analysis-focused *metafor* package (Viechtbauer, 2010) for fitting and testing these models.

Calculating β-diversity

We quantify within site β-diversity as the mean dissimilarity between samples within a site. We first calculate mean Bray-Curtis dissimilarity between samples and then, to standardize β-diversity measures across sites, we normalize mean dissimilarity to a null model. There are many null model algorithms to choose from (Ulrich & Gotelli, 2010). We use the `quasiswap_count` algorithm implemented in *vegan* (Oksanen *et al.*, 2025) which preserves both row and column sums of a sample by species matrix (a "fixed-fixed" type algorithm) and also seeks to maintain matrix fill of permuted matrices close to the observed fill thereby mimicking patchiness in the real data. We choose a fixed-fixed algorithm because we want

to document β-diversity patterns while taking into account possible confounding sampling effects of unevenness in species abundance and unevenness in distribution of individuals across sites. Fixed-fixed algorithms are known to be most conservative in controlling for possible artifactual patterns in species by sample matrix data (Ulrich & Gotelli, 2010). Concern has been raised in the literature that the conservatism of fixed-fixed algorithms may lead to excessive Type II errors [i.e., failure to reject a null hypothesis; Gotelli (2000); Ulrich & Gotelli (2012)], however, our study is not impacted by this possibility because we are not testing a null hypothesis where β-diversity is the response variable.

Within each site we use 500 null sample-by-species matrices to generate null models for each site. The final measure of within site β-diversity is then calculated as a z-score

$$\text{Normalized } \beta = \frac{\text{BC}_{\text{obs}} - \mu_{\text{null}}}{\sigma_{\text{null}}}$$

where BC_{obs} is the observed mean Bray-Curtis dissimilarity between samples within a site, μ_{null} is the mean of the null distribution, and σ_{null} is the standard deviation of the null distribution.

We are also interested in understanding how native and non-native species contribute to the overall mean dissimilarity at a site. We can do this by partitioning Bray-Curtis dissimilarity between native and non-native groups of species. The end result of partitioning Bray-Curtis is two dissimilarity values (one for native species, one for non-natives) that add to the total dissimilarity at a site. Supplementary Section S5.3 provides the mathematical and computational details of how we do this.

We also compare this Bray-Curtis dissimilarity partitioning to a null model. We again use the `quasiswap_count` algorithm but must constrain the algorithm to permute only inside submatrices (one for native species, another for non-natives). This additional constraint is to ensure that the distributions of native and non-native abundances across sites are preserved in permuted and observed site-by-species matrices. We wrote custom code to do this in the *HawaiiArthMETE* package and described in Supplementary Section S5.3.

Comparing observed Bray-Curtis dissimilarity partitions to their permutational null counterparts yields a z-score for the partitioned dissimilarities. While partitions of Bray-Curtis dissimilarity perfectly add to equal the overall dissimilarity, these z-scores will not perfectly add to the previously described overall dissimilarity z-scores because the partitioned z-scores are calculated from a null model with additional constraints (considering native and non-native species) compared to the overall z-scores.

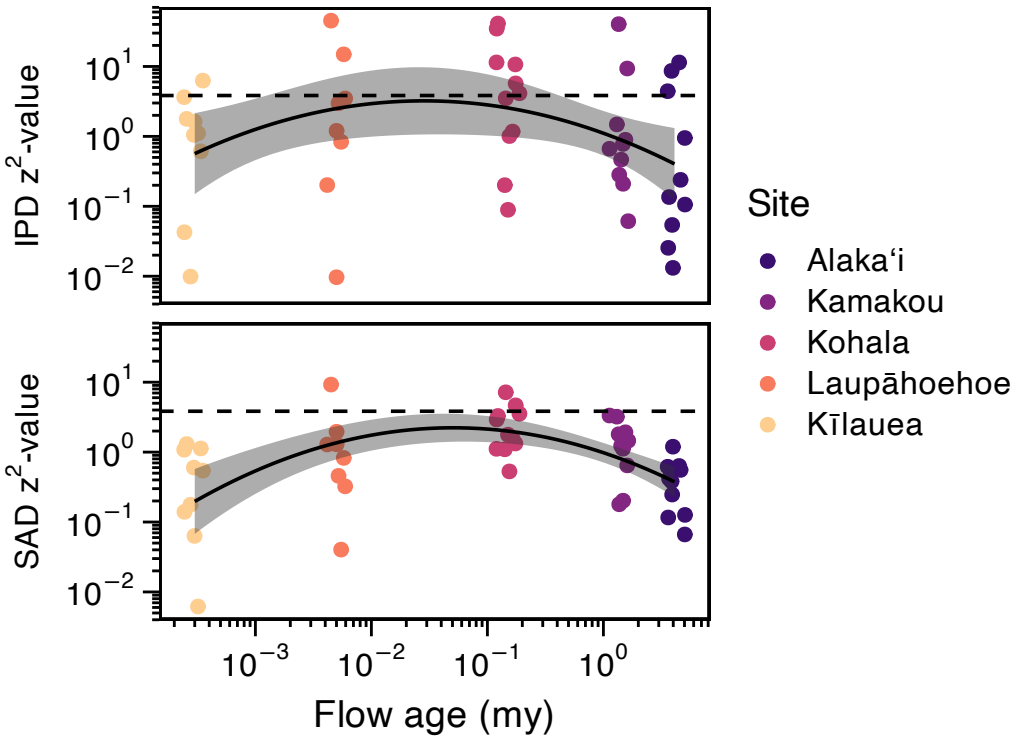


Figure 2: Deviations from METE across the chronosequence as measured by z^2-values. Colors represent sites and match the color coding of substrate age in Figure 1. The horizontal dashed line is the $\alpha = 0.05$ critical value of z^2.

Results

Deviations from METE across the chronosequence

Deviations from METE in the SAD and IPD are largely in agreement: samples with high z^2-values for the SAD tend to have high z^2-values for the IPD (see Supplementary Figure S2). Across all samples this correlation is positive and significant (product moment correlation $r =$ 0.509, \$n = \$ 48, \$P = \$ 0). Within sites, z^2-values of the SAD and IPD are also positively correlated, but only the Laupāhoehoe site has a statistically significant internal correlation (see Supplementary Table 1 and Supplementary Figure S2).

The z^2-values for both distributions peak at middle aged sites along the chronosequence Figure 2. For the SAD this is supported by a likelihood ratio test comparing a quadratic model of z^2-values to an intercept-only model with LRT statistic $\Lambda = 28.73$, $d.f. = 2$, $P < 0.0001$.

The overall spread of IPD z^2-values is greater than that of SAD z^2-values. Not surprisingly, for quadratic model of IPD z^2-values across the chronosequence, the LRT statistic is $\Lambda = 5.513$, $d.f. = 2$, $P = 0.0635$. While the P-value for the test of the quadratic model for IPD z^2-values just misses the ≤ 0.05 significance cut off by the likelihood ratio test, the less conservative Wald test does show the quadratic term to be significant with a 95% confidence interval of [-0.154, -0.014], which does not overlap 0.

Figure 3 shows two exemplar SADs and IPDs with low and high deviation from METE predictions. These exemplars show how METE under-predicts rarity for the SAD and how middle values of metabolic rate can be incorrectly predicted by METE. However, these are only two examples, Figure 4 shows that for the SAD, the primary driver of differences between METE predictions and observed patterns is indeed that observed species abundances have both a higher proportion of singletons and a larger concentration of abundance in the most abundant species compared to METE. Taken together, this means that observed SADs are more uneven than those predicted by METE.

For the IPD there is not a consistent relationship between deviations and the tails of the metabolic rate distribution Figure 4. In terms of proportion of individuals with very low metabolisms, 19 samples showed theory over-predicting this proportion while 27 samples showed theory under-predicting this proportion. The difference was not significant by a binomial generalized linear model of over-versus-under prediction grouped by site (z = -1.173, P = 0.241). In terms of concentration of metabolic rate in the most metabolically active individuals, 24 samples showed theory over-predicting this concentration while 24 samples showed theory under-predicting this concentration. With the two numbers exactly equal, any P-value will never be significant.

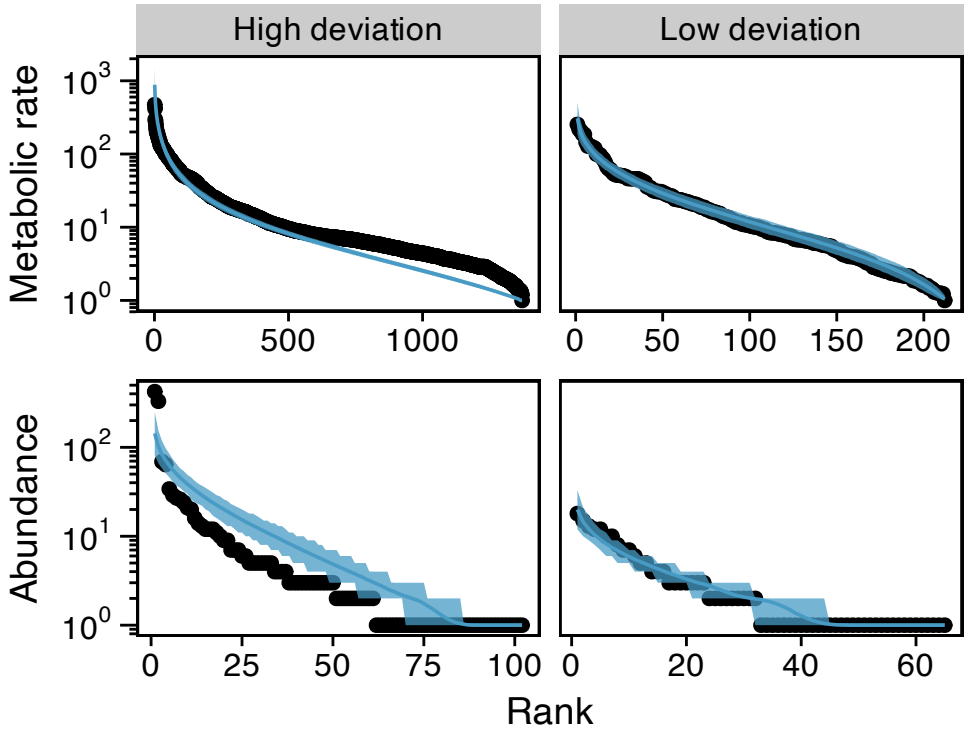


Figure 3: Two exemplar IPDs and SADs showing high and low deviation from METE. The low deviation distributions come from the same sample at the Alakaʻi site while the high deviation distributions come from the same sample at the Kamakou site. The black dots show the observed data while the blue line and shadded blue region show the central tendency and 95% confidence envelopes of 1,000 random draws of data from the SAD and IPD predicted by METE.

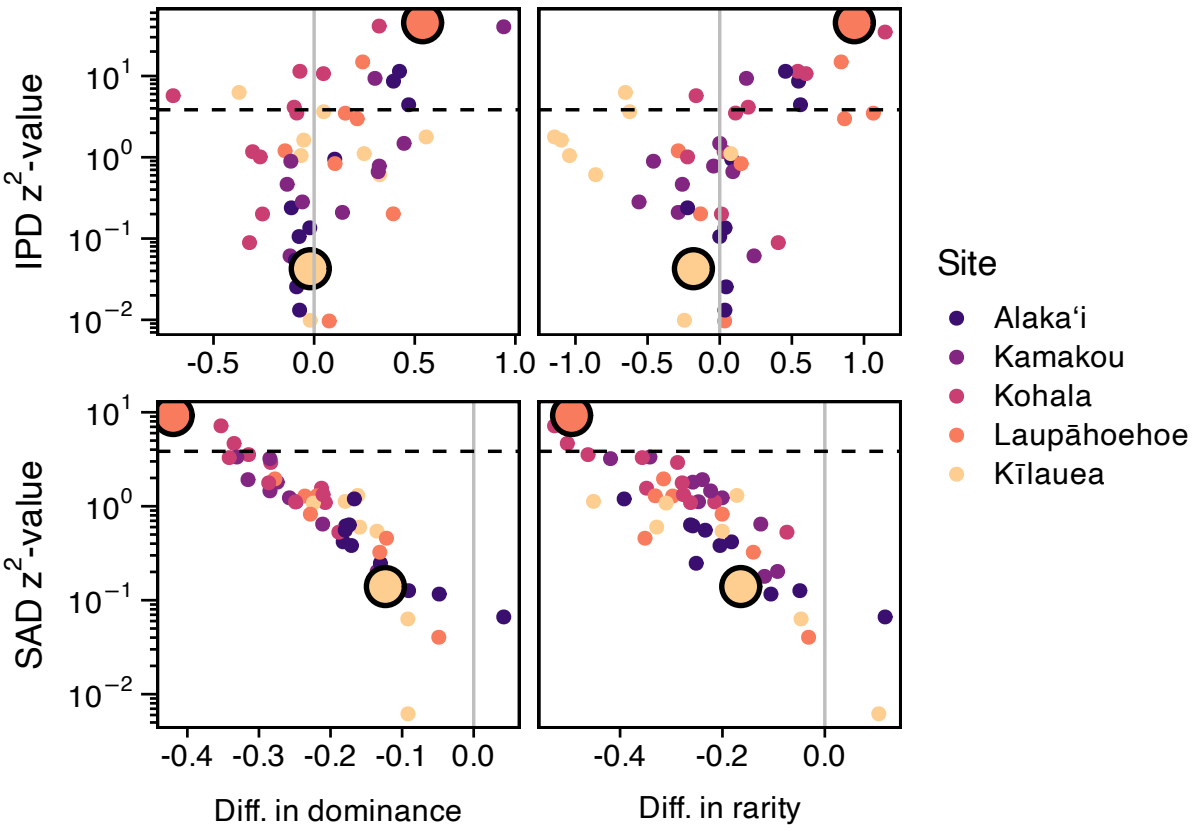


Figure 4: Relationship between z^2 measure of deviation from METE and different metrics describing how observed data and METE predictions diverge. Larger dots correspond to samples used in Figure 3 to exemplify greater and lesser deviation from METE. The horizontal dashed line is the $\alpha = 0.05$ critical value of z^2.

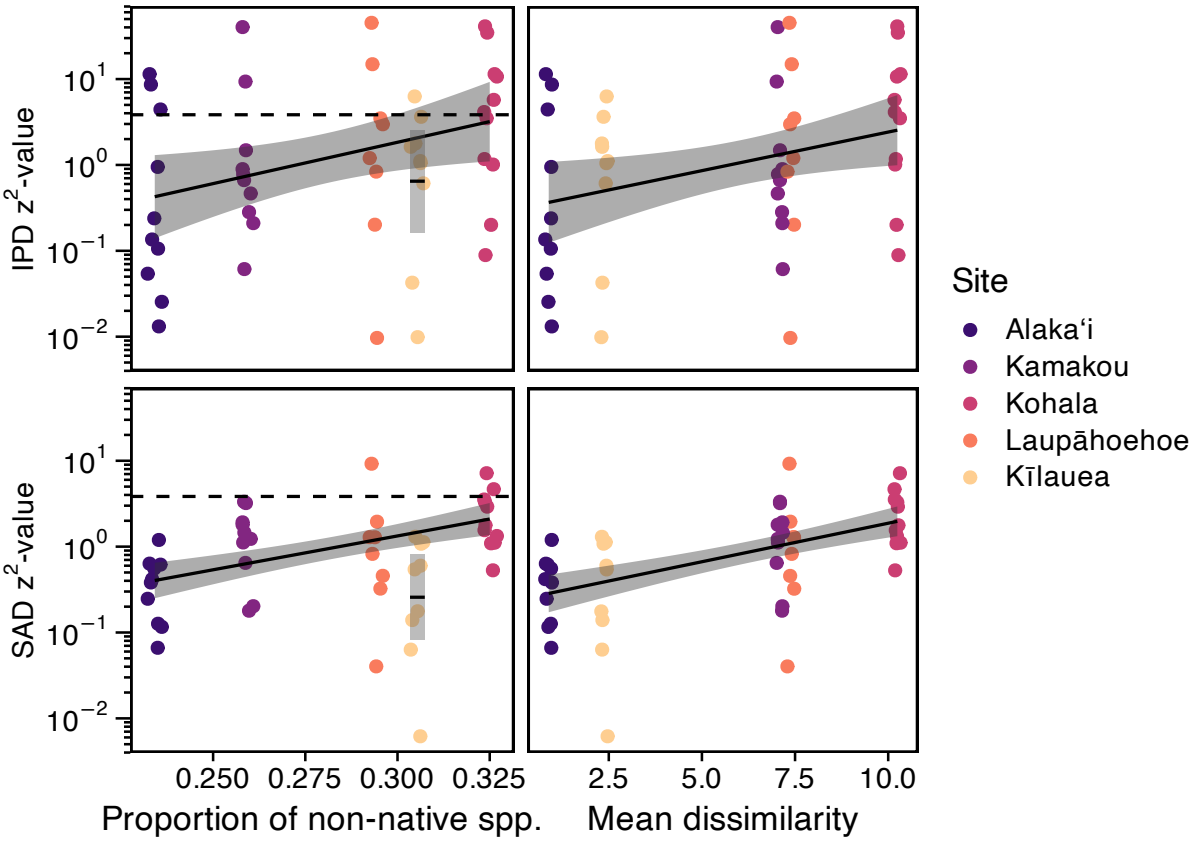


Figure 5: Relationship between z^2 measure of deviation from METE and different predictor variables: proportion of non-native species and β-diversity. Highest β-diversity levels are found at sites of intermediate age, while proportion of non-native species is less consitent across the chronosequence. Horizontal dashed lines are the $\alpha = 0.05$ critical value of z^2.

Deviations from METE predicted by non-native species and β-diversity

The proportion of non-native species at a site has a significant linear relationship with both SAD and IPD z^2-values when excluding Kīlauea Figure 5. For the SAD the LRT of the linear model with positive slope versus the intercept only model is $\Lambda = 22.329$, $d.f. = 1$, $P < 0.0001$. For the IPD the LRT of the linear model with positive slope versus the intercept only model is $\Lambda = 5.262$, $d.f. = 1$, $P = 0.0218$.

Kīlauea is an interesting outlier in both the SAD and IPD z^2-values relationships with proportion non-native species. In the case of the SAD, the 95% confidence interval for the mean of Kīlauea z^2-values is well below the 95% confidence envelope of the overall relationship Figure 5. In the case of the IPD, the 95% confidence interval of the mean of Kīlauea z^2-values is below the 95% confidence envelope of the overall relationship, but the confidence interval and envelope do partially overlap Figure 5. This is a strong suggestion that the effect of non-native species at Kīlauea is distinct from other sites. We unpack that possibility further in our analysis of β-diversity.

Within site β-diversity has a significant positive linear relationship with the z^2-values of both the SAD and IPD. For the SAD the LRT of the linear model with positive slope versus the intercept only model is $\Lambda = 29.968$, $d.f. = 1$, $P < 0.0001$. For the IPD the LRT of the linear model with positive slope versus the intercept only model is $\Lambda = 5.471$, $d.f. = 1$, $P = 0.0193$. There are no clear outlier sites in the relationship of z^2-values and β-diversity.

Partitioning β-diversity across native and non-native species

Kīlauea is an outlier in the relationship between proportion of non-native species and z^2-values but not an outlier in the relationship between β-diversity and z^2-values. To help understand what is different about Kīlauea we partition native and non-native species' contributions to β-diversity. We find that while Kīlauea has a high proportion of non-native species, it also has a low β-diversity, and non-native species contribute negatively, compared to the null, to that low β-diversity Figure 6. Taking these insights together, Kīlauea has many non-native species, but those species are distributed very homogeneously across samples. The oldest site Alakaʻi reverses this pattern: native species contribute negatively to the overall (low) β-diversity at that site.

The oldest (Alakaʻi) and youngest (Kīlauea) sites contrast with Laupāhoehoe, Kohala, and Kamakou sites where both native and non-native species make positive contributions to β-diversity. In Kohala and Kamakou the positive contribution of native species is substantially larger than non-natives, whereas in Laupāhoehoe the contribution of non-natives is larger.

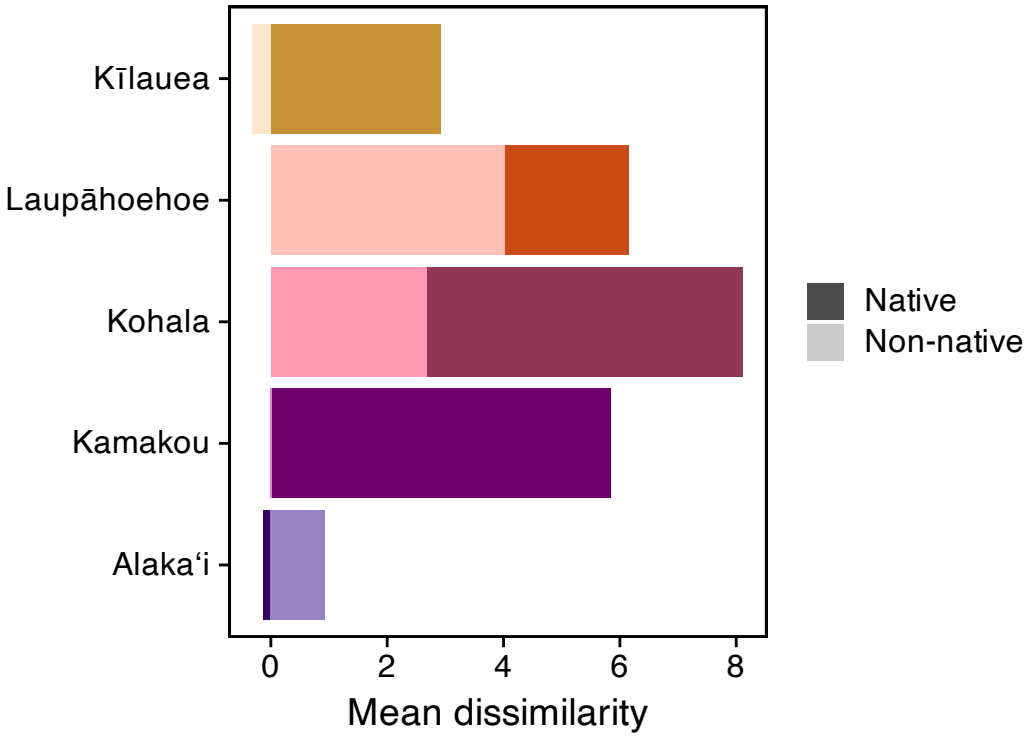


Figure 6: Difference in contribution of native and non-native species to β-diversity compared to null. The total height of bars (taking into account negative heights as well) is equal to the overall β-diversity z-score for a site. These overall β-diversity values are same in pattern, following ahump-shaped curve across geologic age, but different in exact value, to those in Figure 5 because the null model used here is more restrictive (see methods). Lighter colors represent the contribution of non-native species and darker colors the contributions of native species to overall β-diversity. The contribution of non-native species at Kamakou is small and positive (0.02). The contribution of non-native species at Kīlauea is negative and the contribution of native species at Alakaʻi is negative.

Discussion

Real data deviate from the METE-predicted SAD and IPD in consistent ways across the chronosequence, with the greatest departures at middle-aged sites Figure 2. These larger departures are consistent with there being processes not accounted for in the deliberately simple state variables of the theory.

Patterns of deviation in the IPD are more variable than those of the SAD. This is not surprising given that the estimation of metabolic rate from allometric relationships introduces more uncertainty than the direct observation of abundances in samples. The choice to use random samples from allometric relationships, rather than expected values, slightly reduces the variability of IPD patterns because expected values introduce artificial discretization of metabolic rate estimates that are inconsistent with METE and indeed the reality that metabolic rates are continuous (see Supplementary Section S6 and the appendix "Sampling biomass estimates from allometric equations"). Nonetheless, re-running all IPD analyses using expected values from allometric relationships does not substantially change our findings (ssec{sec-re-run}).

Deviations from the IPD are not consistent in any one direction. Theory both under- and over-predicts the shape of the tails of the IPD in roughly equal proportion. Conversely, deviations in the SAD are consistently in the direction of theory under-predicting unevenness Figure 4. In one way, this result is surprising given that METE predicts a log-series distribution (Harte, 2011) which is already one of the most uneven theoretical SAD shapes. However, recent extensive investigation of empirical SADs has found that deviations from statistical baselines akin to METE are often in the direction of more unevenness (Diaz *et al.*, 2021). As pointed out in Diaz *et al.* (2021), deviations toward more unevenness could be accounted for by processes that increase the prevalence of rare species, and/or that increase the abundance of common species. Both niche-based (Chesson, 2000; Yenni *et al.*, 2012) and stochastic dispersal-based (Gotelli, 1991) mechanisms could be valid causes of such processes. Complementary to those possible mechanisms is the possibility that adaptive radiation or elevated neutral speciation could produce both many rare species—young and incipient lineages—and a few highly dominant species—those that have found a successful strategy and/or are more established (Schluter, 2000; Hubbell, 2001; Gillespie & Whittaker, 2025). This possibility becomes relevant in hypothesizing mechanisms for the nature of deviations from METE across the chronosequence.

Across geologic ages, both the proportion of non-native species and β-diversity at a site predict deviations from METE, but in subtly different yet important ways. Across four out of five sites (all but the geologically youngest) deviations from METE increase with the proportion of non-native species Figure 5. The geologically youngest site, Kīlauea, breaks this pattern with a high proportion of non-native species but a small deviation from METE. Across all five sites deviations from METE are consistently predicted by within site β-diversity Figure 5. Both β-diversity and proportion of non-native species point to instability being a driver of deviation from METE.

The fact that geologic age seems to modulate the effect of proportion of non-native species but not within site β-diversity begs the question of how age, biological invasion, and β-diversity relate to each other. β-diversity partitions unequally between native and non-native species across the chronosequence Figure 6. Kīlauea, the youngest site, sees non-native species contributing negatively (compared to the null) to overall β-diversity, leaving native species making the only positive contribution to overall β-diversity. The contributions of native and non-native species to the β-diversity at the oldest site Alakaʻi are reversed. But at both sites overall β-diversity is low. Thus low β-diversity is more consistently associated with low deviation from METE. The high proportion of non-native species at Kīlauea is not consequential because those non-native species are very homogeneously distributed.

The opposite β-diversity contributions of native and non-native species at opposite ends of the chronosequence Figure 6 hints at a mechanistic possibility to explain low β-diversity and low deviation from METE. The assembly of the young Kīlauea site could be dominated by dispersal, which would be consistent with its age—young sites, by definition, have not had sufficient time for diversification and must be assembled by dispersal from source populations (Whittaker *et al.*, 2008). In Hawaiʻi, source populations are typically older volcanoes and islands (Wagner & Funk, 1995). Furthermore, dispersal-driven assembly can have complex impacts on β-diversity including homogenizing it, as with non-native species at Kīlauea, or modestly increasing it if dispersal creates stochasticity in the distributions of species across space (Lu, 2021), as could be the case for native species. Conversely, the oldest Alakaʻi site could represent the outcome of > 4 million years of eco-evolutionary assembly resulting in assemblages that are stable and in steady-state due to niche partitioning and maximized fitness within niches (Chesson, 2000; Godoy *et al.*, 2014; Barabás *et al.*, 2018). This would result in lowest β-diversity among native species, with the rare and patchily distributed non-native species accounting for whatever β-diversity there is. The Alakaʻi site indeed has the lowest proportion of non-native species Figure 5. The Alakaʻi site, located on Kauaʻi, may also experience less dispersal than other islands because there are no older high island than Kauaʻi in the archipelago. While the chronological gap proceeding Kauaʻi leaves a mark in some phylogenies (Price & Clague, 2002) it is unclear if there should be an expectation of less dispersal at the community level.

In between these mechanistic end points of Kīlauea and Alakaʻi, we find both peak deviation from METE at middle-aged sites and large yet variable contributions of both native and non-native to β-diversity Figure 5, Figure 6. These are also the locations along the chronosequence that correspond to the typical minimum time for diversification in Hawaiian arthropods identified by phylogenetic studies (Hormiga *et al.*, 2003; Gillespie, 2004; Bennett & O'Grady, 2013) and consistent with theorized middle-age peaks in diversification (Whittaker *et al.*, 2008). These findings and evidence from the literature are consistent with the hypothesis that young dispersal-assembled communities in an unstable steady state undergo a non-steady state transition via evolutionary diversification to another stable steady state as they age.

The spatiotemporal confluence of these patterns and hypothesized processes along the chronosequence suggests that eco-evolutionary non-steady state might represent an opportunity for biodiversity to change toward a steady state. Left free of anthropogenic pressure, that change

could be achieved slowly by *in situ* diversification. Or, in the presence of anthropogenic pressure, that change could be achieved rapidly through biological invasion. This could explain biological invasion following disturbance (anthropogenic or otherwise) (D'Antonio & Vitousek, 1992; Hobbs & Huenneke, 1992; Burke & Grime, 1996; Davis *et al.*, 2000). It also predicts that the Hawaiian Archipelago and other hotspots of diversification might be particularly prone to biological invasion.

These hypotheses are, however, only suggested by our work and would require further data collection and testing to support. In particular, extensive population genetic study would be needed to confirm the purported history of dispersal at young sites, population adaptation and bifurcation leading to diversification at middle aged sites, and population stability at older sites. New molecular techniques may soon make such study a reality (e.g., Manel *et al.*, 2025).

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